

Problem 2.14

Consider the accumulation functions

$$a_s(t) = 1 + it \quad \text{and} \quad a_c(t) = (1 + i)^t$$

where $i > 0$. Show that for $0 < t < 1$ we have

$$a_c(t) \approx a_s(t).$$

That is,

$$(1 + i)^t \approx 1 + it.$$

Hint: Expand $(1 + i)^t$ as a power series.

Solution

Expand $(1 + i)^t$ as a power series

$$(1 + x)^t = 1 + tx + \frac{t(t-1)}{2!}x^2 + \frac{t(t-1)(t-2)}{3!}x^3 + \dots$$

Substituting $x = i$, we obtain

$$(1 + i)^t = 1 + it + \frac{t(t-1)}{2!}i^2 + \frac{t(t-1)(t-2)}{3!}i^3 + \dots$$

Now, for $0 < t < 1$ and for small values of i , the higher-order terms in $i^2, i^3, \dots$ become negligible. Keeping your marked cancellations to indicate that these terms tend to zero, we write

$$a_c(t) = (1 + i)^t = 1 + it + \frac{\cancel{t(t-1)}i^2}{2!} + \frac{\cancel{t(t-1)(t-2)}i^3}{3!} + \dots$$

Hence,

$$a_c(t) \approx 1 + it = a_s(t).$$

Therefore, under the conditions $0 < t < 1$ and i small,

$$\boxed{a_s(t) \approx a_c(t)}.$$